



MECH 351: THERMODYNAMICS II

Technical Report: Thermoelectric Generator Challenge Project

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EXECUTIVE SUMMARY

This report details the design, simulation and experimental testing of a small vehicle powered by a Peltier thermoelectric generator (TEG). The goal of this project was to further understand the reverse refrigeration, or Carnot cycle by building and analyzing the TEG vehicle's thermodynamic performance and efficiency as it goes up an inclined ramp.

By using a Peltier device with a tea light as the heat source and ice as the cold source, the TEG vehicle was able to climb the ramp at an average velocity of 10.6 mm/s. This represents a 67.88% error difference with the theoretical results outlined in section 4.1. The system's thermal and mechanical efficiencies were found to be 0.217% and 7.23%, respectively.

The result showed discrepancies between the expected theoretical results and the experimental results due to heat transfer inefficiencies and mechanical resistance from the vehicle design, as well as electrical losses from the Peltier device. The report explains these sources of error and shows the key limitations in design. Suggestions for improvement of future designs are explored as well.

This project allowed students to further understand the application of thermoelectric generators and demonstrate the challenges of designing and optimizing an efficient thermal-to-mechanical energy conversion system.

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NOMENCLATURE

h	Height	m
d	Distance	m
$E_{Peltier}$	Energy output from the Peltier device	J
$P_{Peltier}$	Power output from the Peltier device	W
t	Time	s
V	Voltage	V
I	Current	A
$\eta_{Peltier}$	Peltier device efficiency	-
n	Revolutions per minute	RPM
N	Number of teeth	-
ω	Rotational speed	rad/s
v	Linear speed	m/s
r_w	Radius of the wheel	mm
θ	Ramp angle	-
F	Force	N
TEG	Thermo-Electric Generator	-
P_{elec}	Electrical Power Output from TEG	W
Q_{in}	Thermal Energy Input	W
g	Acceleration Due to Gravity	$9.81m/s^2$

1 INTRODUCTION

The objective of this project is to design and construct a small electric vehicle powered by a thermoelectric generator (TEG), also known as a Peltier device, and a DC electric motor. The vehicle is powered by thermal energy provided by a tea light candle or hot water. The generated electric power will be used to propel the car forward across a flat surface and allow it to attempt to climb an inclined ramp. Throughout this project, the design process, testing, and analysis will be documented, with a focus on finding the thermodynamic efficiency of the system.

Thermoelectric generators operate based on the Seebeck effect, where a voltage is generated by a temperature difference across a thermoelectric device. In cooling applications, these devices are electrically powered to create a temperature difference. However, in this project, the Peltier device is used in reverse, as a TEG, where applying heat to one side generates potential difference.

The system functions by creating a temperature difference across the Peltier device. The hot reservoir consists of either a tea light candle or hot water, while the cold reservoir is created by ambient air or cold water. As the temperature difference is applied to the device, a potential difference is produced and is used to drive the car's motor.

Understanding the amount of thermal energy supplied is important when determining the system efficiency. For the tea light, this could involve measuring the mass of wax consumed during operation. When using hot water, the differences between initial and final temperatures can help estimate the energy input. This project bridges the principles of thermodynamics, heat transfer, and electrical power generation, providing insight into the real-world application of thermoelectric systems.

2 PROJECT OVERVIEW

The objective of this project is to further understand thermodynamic power cycles by constructing a small vehicle meant to go up an inclined ramp powered by a Peltier device. The system's overall thermodynamic efficiency will also be analyzed to understand the limitations of a reversed refrigeration cycle.

2.1 Project Requirements

In the context of this project, there are limitations and requirements that must be met.

1. A Peltier device shall be used as the thermoelectric generator
 - a. The Peltier device shall be chosen from a provided list.
2. The heat source shall be a tea light and, optionally, a hot water source.
 - a. The tea light shall only have one wick
3. The DC electric motor may run on 5V or less
 - a. Only 1 DC motor shall be used
4. The ramp shall have an incline of 20° and a length of approximately 3 feet (36 inches)
 - a. The final test shall be made on the ramp provided by the university.
5. The vehicle shall not exceed 10 inches in length
6. All components shall be at ambient pressure.
 - a. There shall be no components carrying material at elevated pressure.
7. The vehicle shall not carry any battery.
 - a. Battery refers to any device storing electrical, kinetic, magnetic, elastic or potential energy.
 - b. Prior to testing, any electrical components on the vehicle shall have an electrical potential of zero.
8. The entire mass of all components shall travel up the ramp
9. The vertical travel shall be measured from starting position to the lowest positioned element at the end of the trial.
10. The vehicle shall be safe.

2.2 List of Components

The table below lists all components used in the project and their role.

Item	Part Number	Quantity	Description
Peltier Device	TEC1-12706	1	Thermoelectric generator from Abra Electronics
DC Electrical Motor	GA12-N20-3V30	1	3V, 30-rpm mini-DC electromechanical motor with metal gears from Abra Electronic
Legos	N/A	a lot	Legos used for base of vehicle, wheels, tea light holder and Peltier device holder
Heat Sink	HS-140	1	40x40x11mm Aluminum Heat Sink for Extruder from Abra Electronics, made of black anodized aluminum
Thermal paste pad	HS-PAD-40	2	Thermal Conductive Silicon Pad, 40mm from Abra Electronics, used to secure the heat sink and copper plate to the Peltier device
Ice	N/A	10g	Cold source
Tea Light	N/A	1	Heat source
Rubber Bands	N/A	3	Used to prevent slip in wheels and secure coke can around heat sink
Copper Plate	N/A	1	Placed between tea light and Peltier device to protect Peltier device from the open flame, very good thermal conductivity
Coke Can	N/A	1	Used as holder for ice on heat sink
Splicing Connector	2946-221-2401-ND	1	Used to reconnect Peltier and motor wires after cutting to measure current while under load

Table 1. List, quantity and description of components used for project

3 PRELIMINARY PLANNING AND TESTING

3.1 Preliminary Calculations

3.1.1 Absolute Height

As the vehicle makes its way up the inclined ramp, the energy provided by the Peltier device will be converted into potential energy.

$$E_{Peltier} = m_v \times g \times h$$

At its maximum output, the TEC-1 12706 Thermoelectric cooler plate purchased from Abra will have a rated voltage of 12V and have a current equal to 6A [1]. The maximum output power of the Peltier device will be:

$$P_{Peltier} = V \times I = (12V)(6A) = 72W$$

Considering that the maximum amount of time allowed to perform the vehicle evaluation is 15 minutes, the Peltier device power can be multiplied by the testing duration to compute the energy output of the device:

$$E_{Peltier} = P_{Peltier} \times t = (72W) \left(15min \times \frac{60s}{1min} \right) = 64,800J$$

Experimentations on Peltier devices have shown them to have an efficiency of around 2% for a corresponding temperature gradient of $55^{\circ}C$ which is close to the maximal temperature difference for this Peltier device [2]. Applying this efficiency to the energy output of the Peltier device gives:

$$E_{Peltier} = \eta_{Peltier} \times E_{Peltier} = (0.02)(64800J) = 1296J$$

If it's assumed that the vehicle weighs 150g which is around the weight of the vehicle designed for this project, then the maximum absolute height is:

$$h = \frac{1296J}{(0.15kg)(9.81m/s^2)} = 880.73m$$

Under real-life conditions, all the Peltier device energy would not be able to be converted into useable power to climb the ramp. Frictional losses from the wheel friction and limitations from the motor's maximal work output would limit the actual highest reachable height.

3.1.2 Expected Maximum Velocity

The GA12-N20-3V30 mini-DC motor purchased from Abra is specified to have a speed of 25 RPM when it is at maximum efficiency which occurs at the rated voltage equaling 3V [3]. Considering that a 12 teeth bevel gear was connected to the motor shaft for power transmission to a 20 teeth bevel gear, the speed will be reduced.

$$n_{out} = n_{in} \times \frac{N_{in}}{N_{out}} = (25RPM) \left(\frac{12 \text{ teeth}}{20 \text{ teeth}} \right) = 15RPM$$

Converting from RPM to angular speed:

$$\omega = \frac{15rev}{min} \times \frac{1min}{60s} \times \frac{2\pi rad}{1rev} = 1.5708rad/s$$

Then, knowing that the radius of the Lego wheels measures 37mm, the maximum ideal velocity is:

$$v = \omega \times r_w$$

$$v = (1.5708rad/s)(37mm) = 58.12mm/s$$

However, this velocity value of 58.12mm/s is the velocity if the voltage obtained was equal to the rated voltage of 3V. The actual maximum voltage achieved from our Peltier device setup was 1.74V. Accounting for this difference the actual expected velocity can be calculated. Knowing the speed is directly proportional to the voltage [4]:

$$\omega_{actual} = \omega_{ideal} \times \frac{V_{actual}}{V_{rated}} = (1.5708rad/s) \left(\frac{1.74V}{3V} \right) = 0.9111rad/s$$

$$v_{actual} = \omega_{actual} \times r_w = (0.9111rad/s)(37mm) = 33.3107mm/s$$

3.1.3 Expected Height

Applying the same procedure as in Section 3.1.1, the height can be found by using the experimentally obtained values for the voltage, current, and vehicle mass.

$$P_{Peltier(actual)} = V_{actual} \times I_{actual} = (1.74V)(0.046A) = 0.08004W$$

Multiplying it by the vehicle evaluation duration will give the energy output from the Peltier device:

$$E_{Peltier} = P_{Peltier} \times t = (0.08004W) \left(15min \times \frac{60s}{1min} \right) = 72.036J$$

Using the final fully equipped vehicle weight of 177g and the Peltier device efficiency, the maximum absolute height is calculated to be:

$$h_{actual} = \frac{E_{Peltier} \times \eta_{Peltier}}{m_v \times g} = \frac{(72.036J)(0.02)}{(0.177kg)(9.81m/s^2)} = 0.8297m$$

Knowing that the height of the ramp that must be climbed for this project is equal to 0.3127m:

$$h_{ramp} = d \times \sin\theta = (0.9144m)(\sin 20^\circ) = 0.3127m$$

$$0.8297m > 0.3127m$$

Therefore, the vehicle will be capable of successfully climbing the incline ramp.

3.2 Experimental Testing of Peltier Voltage Output

3.2.1 Version 1

3.2.1.1 Design

For version 1, our setup to generate as much power as possible from the Peltier device can be seen in Figure 1.

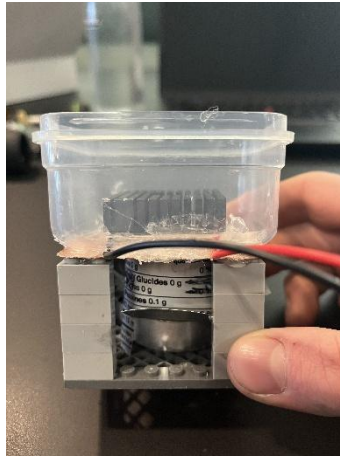


Figure 1. Front view of version 2 of the thermoelectric generator system

Using the Peltier device as the main structure to build off of, we stuck a thermal pad on both sides of the device. On one side we attached a copper plate to evenly distribute the heat from the candle, and on the other side we attached the metal heat sink. On the side with the heat sink we cut a hole in the bottom of a plastic Tupperware container, placed it over the heat sink and sealed any gaps with hot glue. This container served to hold the ice and water on the cold side the that the heat sink had as much surface area being cooled as possible. On the side with the candle and copper plate, we used a cut up aluminum soda can to funnel the heat from the candle straight up to the copper plate while allowing air to flow through the top and bottom so as not to suffocate the candle.

3.2.1.2 Results

Once version 1 was constructed, we lit the candle, filled the Tupperware with ice and water, and attached a voltmeter to the ends of the Peltier wires to measure our voltage output. Starting from zero, our voltage output climbed as the temperature difference between the two sides of the Peltier device increased and stabilized. After giving sufficient time for the temperature difference to maximize, we measured a peak voltage of approximately 1.55V.

3.2.2 Version 2

3.2.2.1 Design

Although we reached sufficient voltage to power our car with version 1, we chose to redesign the generator portion of our car to try to minimize weight as our initial solution was relatively heavy using the Tupperware setup. This redesign can be seen in Figure 2.

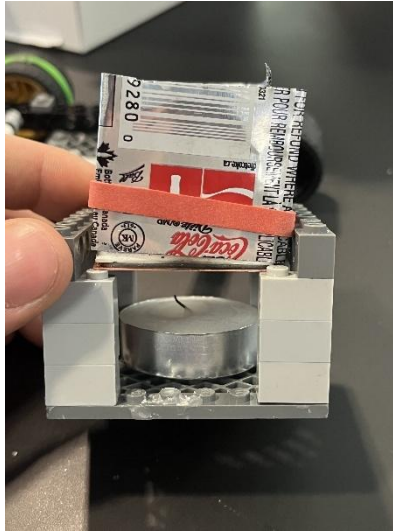


Figure 2. Front view of version 2 of the thermoelectric generator system

For version 2, we utilized the same base system consisting of the Peltier device, thermal paste pads, copper plate, and heat sink, but made some changes from there. Firstly, we chose to remove the aluminum can funnel from the lower portion of our generator. Instead, we elected to rely on the Lego structure supporting the Peltier device to also direct as much of the candles heat as possible towards the copper plate. This solution increased airflow around the candle and eliminated unnecessary weight as we could have one element that served two purposes. Secondly, we exchanged the watertight Tupperware container on top for a structure made from an aluminum pop can. Secured with an elastic band, this structure held the ice in place through the trial and allowed excess water to drip away through the trial as the ice melted thus eliminating excess weight as the vehicle progressed along the course.

3.2.2.2 Results

This version two exceeded our expectations as not only did it weigh significantly less than our original solution, but it performed better as well. When we ran a trial with our new system, the measure voltage reached a new and improved maximum of 1.74 volts.

3.3 Vehicle Construction & Performance Testing

To focus on the thermal system, the team decided to use LEGOs as the base for the vehicle. There are no custom parts that were printed or manufactured for this project, except for a soda can used as the ice recipient for the cold side of the Peltier device.

3.3.1 Design Iterations

3.3.1.1 Iteration 1

The first model of the vehicle chassis was designed prior to procuring the Peltier device, the motor, and other components. Therefore, no testing had taken place to determine certain design constraints such as the impact of vehicle weight. It was simply created as a starting point to have a working model once the components were purchased. Constructed out of Lego parts, it was 24.4cm in length, 3.2cm above the ground from the bottom of the chassis and used 4 Lego wheels with a diameter of 81.6mm. Additionally, the entire frame weighed 207g. The gear configuration was a set of bevel gears with a 12-tooth gear planning to be connected to the motor shaft in contact with a 36-tooth gear on the axle.

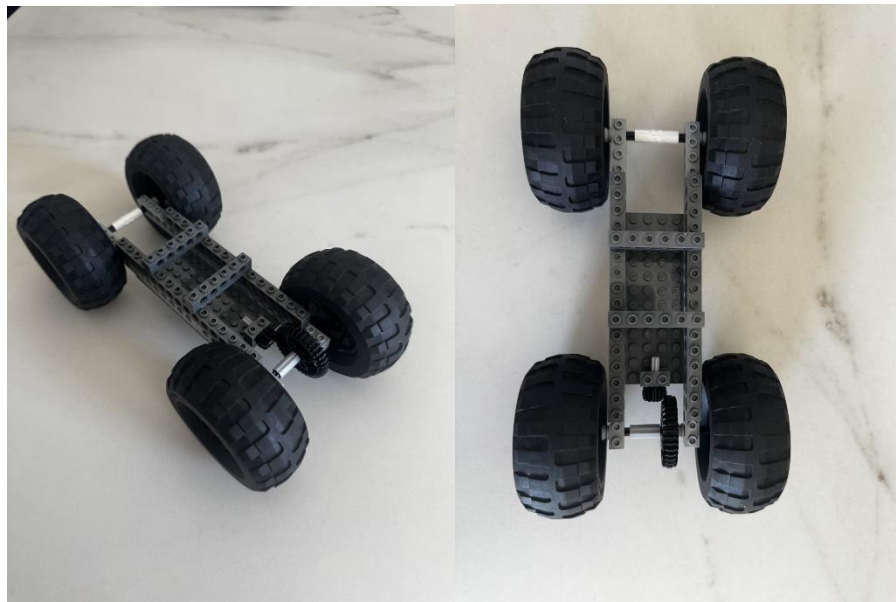


Figure 3. Isometric and top view of the first designed model of vehicle chassis

Once the components were purchased, the chassis and motor were finally seen together, and it became apparent that the model would be too large to use with the motor. It would have been

difficult for the small motor to be able to function under the weight that did not even include the Peltier device and other structural elements to hold everything together.

3.3.1.2 *Iteration 2*

The second iteration of the vehicle chassis had the goal of greatly diminishing the weight of the vehicle and eliminating redundant components. Firstly, the wheels were changed to Lego wheels with a diameter of 37mm instead of 81.6mm, and 3 wheels were used instead of 4. This was done to eliminate unnecessary weight since the vehicle would still be well balanced with one less wheel. The chassis was simplified to include less Lego parts and include compartments to hold the motor, candle, and Peltier device. This second model measured 16.8cm in length, the bottom of the chassis is 0.9cm above the ground and including the motor the vehicle weighs 126g.

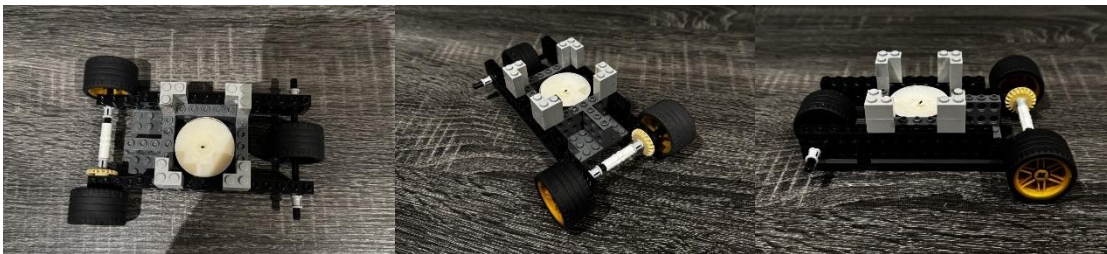


Figure 4. Top, isometric and side view of the second iteration of the vehicle design

The vehicle was able to lose 81g of weight, however when accounting for the weight of the other components needed to be added it would weigh approximately 180g. The gear ratio was also changed because the gears used in the first model were too large. A 12-tooth bevel gear connected to the motor shaft would transmit power to a 20-tooth bevel gear connected to the front-wheel axle. A speed reduction of 0.6 would occur from this gear configuration which would inversely increase the torque by a ratio of 1.67. An increase in torque was desired because the ability of the motor to carry the vehicle up the inclined ramp is more important than its speed.

3.3.1.3 *Iteration 3*

The third iteration of the chassis optimized the space for the components and certain Lego parts were changed to decrease the weight without affecting the structural integrity of the frame. For example, Lego plates that were previously used to hold the vehicle together, were changed for plates with a mesh texture because less plastic was used. Additionally, rather than having a compartment separated by Lego bricks to hold the motor, using the mesh plate a zip tie could be

used to hold the motor in place. The wires of the Peltier device were also cut to a shorter length to reduce the weight and avoid entanglements with other components before being soldered to the motor. The same gear configuration as the second iteration was used.

To connect the 12-tooth Lego gear to the motor shaft, hockey tape was initially used to increase the diameter of the shaft to create a tight fit between a tube holding the gear to the motor shaft. However, after a few trials with the motor running, the tape would become loose, and the tube would rotate separately from the motor shaft. Instead, super glue was used to fix the gear to the motor shaft.

The vehicle measurements for the third iteration were 14.8cm in length, 11.3cm in width, 0.9cm above the ground. The total weight including the candle and Peltier device was 157g.

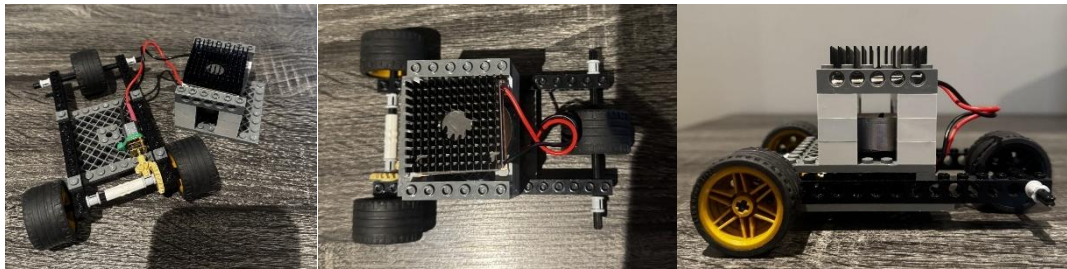


Figure 5. Top and side view of the third vehicle iteration

With all the components integrated into the vehicle, numerous test trials were performed to evaluate its performance. The vehicle had no issue driving on the flat surface, but when climbing up the ramp it would lose traction and be unable to go climb any further. Therefore, the friction between the tires and the wooden surface of the ramp needed to be increased.

3.3.2 Final Vehicle Design

The final design of the vehicle is structurally the same as the third iteration, however certain components were added based on the observations from testing the third model. To improve the traction of the front wheels when climbing up the ramp, elastic bands were added to both tires. This proved to be sufficient to allow the vehicle to climb to the top of the ramp without any slippage or stalling.

Wanting to insulate the ice reservoir to maintain its cold temperature, it was decided to surround the outside of the Peltier device with a thin sheet of aluminum from a soda can. Held

together by elastic bands, the aluminum walls were able to hold more ice and limit the interaction with the warmer ambient room temperature air. The aluminum walls were not sealed at the corners between the Peltier device which allowed for the melted ice water to drip off the heatsink. This was seen as a good thing because the water would drip off the vehicle which would reduce the vehicle weight. However, some of the water ended up landing on the motor which could damage it. So, a thin layer of Saran wrap was placed over the motor to ensure no water could come in contact with it.

The total weight of the final vehicle was 161g with all its components except for the ice. Adding 3 ice cubes increased the weight by 22g which gives a total weight of 183g. Just like the third iteration the final vehicle has a length of 14.8cm, a width of 11.3cm, a total height of 9.6cm, and the bottom of the chassis is 0.9cm above the ground.

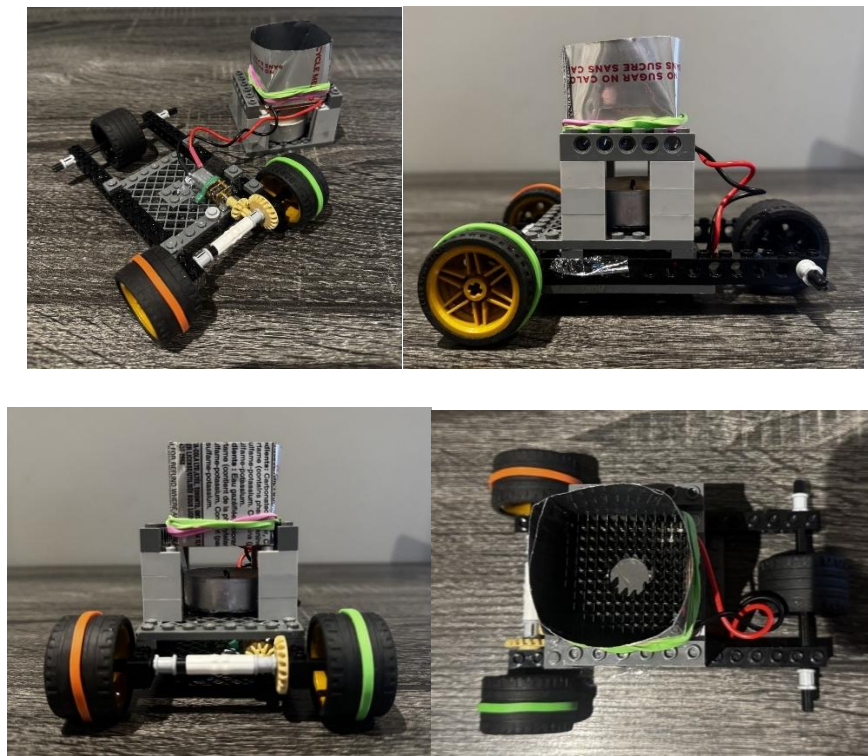


Figure 6. Pictures of the final thermoelectric car design

3.3.3 Measured Performance Data

The collected vehicle test data from our final design can be seen below in Table 2

	Flat	Inclined	Full Run
Voltage (V)	1.74	1.74	1.74
Maximum Current (mA)	26.6	46.0	46.0
Hot Side Temp (°C)	25.4	25.7	25.4
Cold Side Temp (°C)	4.1	4.2	4.1
Displacement (m)	0.3	0.3	1.85
Time (s)	17.2	35.45	173.8
Velocity (m/s)	0.0174	0.0085	0.0106
Mass of vehicle (g)	177		

Table 2. Vehicle test data

This table denotes all parameters of the vehicle that were measured to quantify our designs performance. Voltage was found by measuring the maximum output of our thermoelectric generator setup as discussed in Section 3.2. To measure the maximum current throughout operation of our vehicle, an ammeter was attached in series between our motor and our Peltier device. Readings were then taken on the flat and inclined sections of the course individually. The setup for such measurements can be seen below in Figure 7.

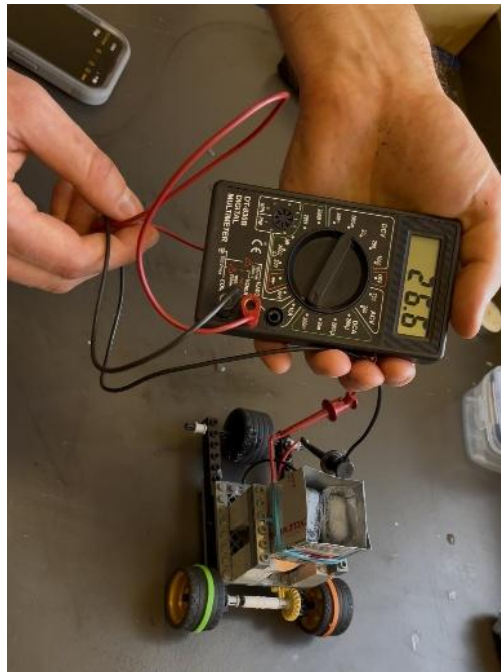


Figure 7. Experimental setup to measure current

Hot and cold side temperatures were measured immediately after each trial using an infrared thermometer. The temperature was measured at the copper plate for the hot side, and the heat sink for the cold side. To calculate the speed of our vehicle at different points of the trial, a measuring tape was set up, and the amount of time needed to travel a certain distance was measured. Using distance and time information, linear speed was calculated using the formula $v = \frac{d}{t}$. Finally, the mass of the loaded vehicle was measured using a scale.

4 DISCUSSION

4.1 Expected vs. Actual Results

The comparison between expected and actual results serves as an indication as to whether the assumptions and calculations made in the theoretical calculations were aligned with reality and can help make better and more informed decisions in future. Regarding the predictions for the thermodynamic car project, the theoretical and experimental results fall within the order of magnitude of each other and successfully predicted the ability of our design to complete the trial. For linear speed, the expected velocity of our solution was approximately 33.3mm/s while the actual measured velocity was found to be 10.6mm/s. This represents a 67.88% error difference, however, the actual speed falling below the expected speed makes sense as no other forces were considered such as resistance to rotation in the system, slippage, etc. Additionally, the motor was assumed to be operating under maximum efficiency under proper loads. Thus, the results differing by a factor of approximately 3x aligns with expectations in the situation. As for the predicted height the vehicle could reach, the expectation was that our solution could reach a maximum height of 0.83m in 15 minutes. While the maximum height our vehicle could attain was not directly tested, our vehicle managed to reach the top of the ramp within the time allotted for the trial thus verifying the theoretical prediction of a successful solution. Although an extrapolation of the time it took our vehicle to climb the ramp suggests that our design could have climbed higher than the 0.83m in the allotted time, again, the expected and actual results falling within the same order of magnitude of each other is promising, considering the simplifying assumptions made.

4.2 Efficiency Analysis

4.2.1 Thermodynamic Efficiency

The overall thermodynamic efficiency is measured using the standard thermal efficiency formula:

$$\eta_{over} = \frac{W_{mech}}{Q_{in}}$$

W_{mech} is found using the formula:

$$W_{mech} = mgd\sin\theta$$

Where the mass of the vehicle is equal to 0.162 kg, the θ is 20° , and the d is 1.85m, which is the total distance of the test.

The Q_{in} was found using the equation:

$$Q_{in} = \Delta m_{wax} h_{wax}$$

Where Δm_{wax} is the weight of the burnt wax was found to be 0.11g; h_{wax} is the heat combustion of wax and is 42MJ/kg.

Putting these numbers into the equation gives an overall thermal efficiency of:

$$\eta_{over} = \frac{0.162(9.81)(1.85) \sin 20}{0.00011(42E + 06)} \cdot 100\% = 0.217\%$$

This result suggests that the system has significant inefficiencies, likely due to heat losses, inefficiencies in energy conversion, and friction.

4.2.2 Mechanical Efficiency

The mechanical efficiency is found using the equation:

$$\eta_{mech} = \frac{W_{mech}}{P_{elec}t}$$

Where the W_{mech} is the same as found in section 4.2.1, P_{elec} is 0.08W, and t is the total distance travelled during testing, which is equal to 173.8 seconds. Plugging this data into the equations gives.

$$\eta_{mech} = \frac{1.005}{0.08(173.8)} \cdot 100\% = 7.23\%$$

This suggests that while the motor is relatively efficient, a significant portion of electrical power is lost to friction, air resistance, and drivetrain inefficiencies.

4.2.3 Thermodynamic Efficiency of Peltier Device

The thermodynamic efficiency of the Peltier device is found using the formula:

$$\eta_{TEG} = \frac{P_{elec}}{Q_{in}}$$

Where Q_{in} is the same value found in section 4.2.1 and P_{elec} is the same value found in section 4.2.2. Plugging the values into the equation gives:

$$\eta_{TEG} = \frac{0.08}{4620} \cdot 100\% = 0.0017\%$$

This extremely low efficiency reflects the inherent limitations of thermoelectric conversion, particularly in small-scale applications where heat dissipation and material inefficiencies play a significant role.

4.3 Design Limitations and Sources of Error

The main challenge of this project was to build an efficient thermal-to-mechanical energy conversion system with the requirements described in Section 2.1. Due to the limitations in the heat source and the size of the DC motor, the vehicle had to be designed to be small and lightweight, but heavy enough to have resistance to go up the ramp.

4.3.1 Heat Transfer Inefficiencies

The team's main concern was heating transfer inefficiencies due to the tea light recipient. If the Peltier device was too far away from the tea light, there would be too much heat loss due to the heat dissipating in the surrounding air through radiation and convection. This was seen in the first iteration, where the tea light was held around 4 centimeters away from the Peltier and the maximum voltage obtained was around 0.5 V. Following this initial test, a soda can was cut and placed around the light so that the heat would build up and reduce heat loss. However, this only

gave around 0.8 V to the Peltier device. Finally, the tea light height was changed to around 2 centimeters and the Peltier's voltage reached 1.5 V with ice as the cold source.

Another concern for heat loss was due to the ice recipient. In one of the initial design iterations, a small plastic container was used and sealed to the Peltier using hot glue. This was not tightly sealed and allowed water to seep through. The shape of the container also meant that the heat from the flame was in contact with the ice recipient and melted the ice more rapidly, leading to a smaller heat difference in the Peltier. This was fixed by changing the recipient to a smaller aluminum one. However, this still leaves room for inefficient and inconsistent cooling from the cold side.

It is also important to consider the variations of the flame height and heat output between each test. This can be due to the wick length, amount of wax left and small differences in the tea light placement. This may have led to differences in the testing results.

4.3.2 *Electrical Losses*

There may also be discrepancies due to the Peltier device and the DC motor. In the case of the Peltier, its internal resistance can lead to energy losses, which would affect the voltage output. This is also the case for the DC motor. Moreover, loose wiring between the Peltier and the DC motor may have led to variations in the electrical system's resistance. This would lead to less power being delivered to the DC motor, thus reducing the velocity output of the vehicle.

4.3.3 *Mechanical Losses*

The vehicle build and the choice of materials may also have led to system inefficiencies. The “gear box” used in the system is made of LEGOs, which may not have the best meshing or alignment for gears. This could lead to slipping between the gears, reducing the overall mechanical efficiency of the system. There is also rolling resistance and friction due to the vehicle weight and wheel material. Although some friction is necessary to prevent slipping on the inclined ramp, too much resistance may reduce the overall speed and mechanical efficiency of the vehicle. Finally, if the wheels are not perfectly aligned, there may be losses due to poor motion transfer.

4.4 Future Improvements

There are many improvements that can be made to increase the overall thermal and mechanical efficiency of the system.

As discussed in Section 4.3.1, the way the tea light is set up greatly impacts the heat transfer efficiency to the Peltier device. One way to improve in the design would be to make an isolated chamber to reduce heat losses to the surrounding air, while still maintaining air flow so the candle can burn. Moreover, this would allow the cold side to be protected from any heat coming from the tea light, thus stabilizing the temperature difference between both sides of the Peltier device. Moreover, it would be beneficial to improve the cooling efficiency of the cold side. Although a heat sink has been added, introducing a fan or a larger and more thermally conductive ice recipient would increase the cooling efficiency further. This could also be a circulating-water system rather than the simply putting ice on top of the heat sink.

The vehicle itself can also be optimized to reduce mechanical and electrical losses. First, proper soldering and wire attachment would ensure that there is less resistance and higher power delivery to the motor. Further testing and comparison of various Peltier devices would also be pertinent to choose the most efficient one. When building the rotational-to-translation motion system, choosing gears of higher precision for better meshing can lead to better mechanical output. The vehicle's wheel can also be aligned more precisely to ensure there is no mechanical unbalance and rolling resistance. It would also be pertinent to try to reduce the overall weight of the system by changing the materials and vehicle size.

5 CONCLUSION

The goal of this project was to use a Peltier device to power a mechanical solution to the thermoelectric car challenge presented in the course MECH 351, to further understand thermodynamic systems. Through many difficulties and design iterations, in the end, our solution successfully completed the challenge by traveling the length of the course and up the ramp within the time allotted. It managed to produce a peak voltage and current of 1.74V and 46.0mA and achieved a maximum velocity of 17.4mm/s on the flat portion of the course and 10.6mm/s overall velocity on the flat portion and up the ramp. Although not perfect given the simplifying assumptions, our theoretical predictions of the vehicle's behavior aligned with the actual result and accurately predicted the successful completion of the trial. Despite the inherent inefficiencies in powering a vehicle using a Peltier device, a future design could nonetheless be improved by further optimizing the temperature differential across the device, reducing mechanical losses due to friction, and reducing the overall weight of the solution. This project served to challenge and broadened students' knowledge of thermodynamic principles and allowed for an opportunity to hone design and manufacturing skills.

6 REFERENCES

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